

Scalability & Future Plan

Date	14 July 2026
Team ID	
Project Name	HDI Predictor (Human Development Index Predictor)
Team Size	1 (Individual Project)
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority
1	SQLite history resets on every Render redeploy (ephemeral filesystem)	Users lose custom prediction history across deployments	Medium
2	No user authentication — history is shared/global, not per-user	Not suitable for multi-user production use as-is	Medium
3	Free-tier hosting sleeps after inactivity	First request after idle time takes 10–30 seconds to respond	Low

Scalability Plan

S.No	Aspect	Current State	Proposed Upgrade
1	User Load	Single Gunicorn worker, free tier	Add more workers / paid tier with autoscaling as traffic grows
2	Data Storage	SQLite (file-based, ephemeral on Render free tier)	Migrate to managed Postgres for durable, concurrent-safe storage
3	Performance	Model loaded once per process (cached)	Add response caching for repeated identical Home-page queries
4	Security	No auth; all endpoints public	Add per-user auth and rate-limiting on the prediction API

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Migrate predictions.db to managed Postgres	Next iteration	Prediction history persists permanently across redeloys
Phase 2	Add lightweight user accounts	Next iteration	Each user gets a private prediction history
Phase 3	Add year-over-year HDI trend prediction (time series)	Future	Supports forecasting, not just single-point prediction